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Regional Variation in Guttae Distribution in Fuchs Endothelial Corneal Dystrophy

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Purpose: To investigate regional variations in guttae distribution in Fuchs endothelial corneal dystrophy (FECD) and evaluate its relationship with anatomical ultraviolet (UV) protection by the eyelid.

Design: A prospective observational study.

Participants: This study involved 19 eyes of 19 patients with FECD between April 2022 and June 2024 at Kyoto Prefectural University of Medicine.

Methods: Using slit-scanning wide-field contact specular microscopy, we analyzed guttae distribution in 5 corneal regions (central, superior, temporal, nasal, and inferior). Marginal reflex distance (MRD) was measured to assess the eyelid position. Proportion of guttae was quantified using image analysis.

Main Outcome Measures: Regional differences in guttae distribution and correlation with MRD measurements.

Results: The superior region demonstrated a significantly lower proportion of guttae (29.4%) compared with central (69.0%), temporal (57.4%), and nasal areas (57.5%) (all $P < 0.05$). The difference between superior and inferior areas (45.6%) was not significant ($P = 0.08$). Margin reflex distance measurements showed no significant correlation with superior guttae distribution ($P = 0.40$).

Conclusions: Our study reveals a distinct regional variation in FECD progression, with relative sparing of the superior cornea suggesting potential UV-protective mechanisms. Although not directly correlated with eyelid position, these findings indicate that environmental factors may influence disease progression. This insight suggests the potential benefit of UV protection as a preventive strategy, particularly in early-stage FECD patients.

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Fuchs endothelial corneal dystrophy (FECD) is a progressive and multifactorial disorder characterized by the formation of guttae and irreversible damage to corneal endothelial cells (CECs), leading to corneal edema.[1,2] In degenerated CECs, endoplasmic reticulum stress induces the unfolded protein response, activating the intrinsic apoptotic pathways of these cells. Furthermore, these degenerated CECs often undergo epithelial–mesenchymal transition, leading to the accumulation of abnormal extracellular matrix in Descemet membrane.[3]

Fuchs endothelial corneal dystrophy is a complex and heterogeneous genetic disorder.[1] The strongest genetic association with FECD is with the cytosine-thymine-guanine repeat expansion within the third intron of the transcription factor 4 (*TCF4*) gene,[4,5] but other common genetic mutations such as *COL8A2*, *SLC4A11*, *ZEB1*, and *ABGL1* have also been reported.[6] Beyond these genetic factors, environmental factors such as smoking and ultraviolet (UV) exposure are recognized as additional significant risk factors for FECD development, suggesting a synergistic interaction between genetic and environmental factors.[7–9]

In the early stages of FECD, guttae initially appear in the central cornea and later spread to the peripheral areas.[10,11] Studies have documented varied distribution patterns—some reported horizontal progression from the central or paracentral areas.[12,13] In contrast, others noted higher densities of the inferotemporal area with relative sparing of the nasal and superior areas.[14] However, the mechanisms underlying these regional differences remain poorly understood. While anatomical protection by the eyelid is hypothesized to reduce UV exposure in the superior cornea, direct evidence linking eyelid position to guttae distribution is lacking.

Slit-scanning wide-field contact specular microscopy facilitates the acquisition of higher-resolution images of guttae, allowing for more detailed analysis and evaluation compared with noncontact specular microscopy when corneal edema is not pronounced.[15,16] Our group has previously reported extraction and quantification of guttae images in patients with FECD obtained through slit-scanning wide-field contact specular microscopy.[17]

This study aimed to utilize slit-scanning wide-field contact specular microscopy to comprehensively analyze the

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regional distribution of guttae, with a particular focus on investigating the potential protective role of the eyelid against UV exposure.

Methods

Patients

This prospective observational study was approved by the Ethics Committee of Kyoto Prefectural University of Medicine, Kyoto, Japan, and was conducted in accordance with the principles of the Declaration of Helsinki. From April 2022 to June 2024, 5 corneal specialists (Y.T., K.K., H.F., C.S., and S.K.) diagnosed FECD based on the presence of bilateral corneal guttae. Patients were enrolled after obtaining informed consent. Patients with corneal edema, those unable to undergo contact specular microscopy examination, a history of intraocular surgery, contact lens wear, or other corneal disorders were excluded. Disease severity was classified according to the modified Krachmer grading scale (Fig 1A, B).

Calculation of Margin Reflex Distance

The marginal reflex distance (MRD) was measured using Idra (Inami, Japan). Patients were instructed to gaze straight ahead while photographs were taken. Marginal reflex distance was measured as the vertical distance from the corneal light reflex to the upper eyelid margin (Fig 1C).

Creation of Panoramic Images

Corneal endothelial cell images were obtained using a slit-scanning wide-field contact specular microscope (CellChek; Konan Medical). Panoramic images of CEC were created from video footage, excluding frames captured during eye movement or those that did not clearly visualize CECs and guttae. Any method to enhance image quality during extraction from the video was not applied. Initially, the central area was imaged, followed by imaging from left to right and from bottom to top (Fig 2A). The right eye was selected; if the right eye had undergone any surgery, the left eye was chosen instead.

CEC Image Analysis from the 5 Different Regions

Using a slit-scanning wide-field contact specular microscope, panoramic images were created with a diameter of 6 mm. Corneal endothelial cell images were extracted from the central area and 4 peripheral regions (superior, temporal, nasal, and inferior areas, each 3 mm from the center) (Fig 2B–F). We performed guttae extraction and quantification using methods previously reported by our group.[17] Briefly, quantitative image analysis was performed using ImageJ/Fiji software[18] with customized macros for the extraction

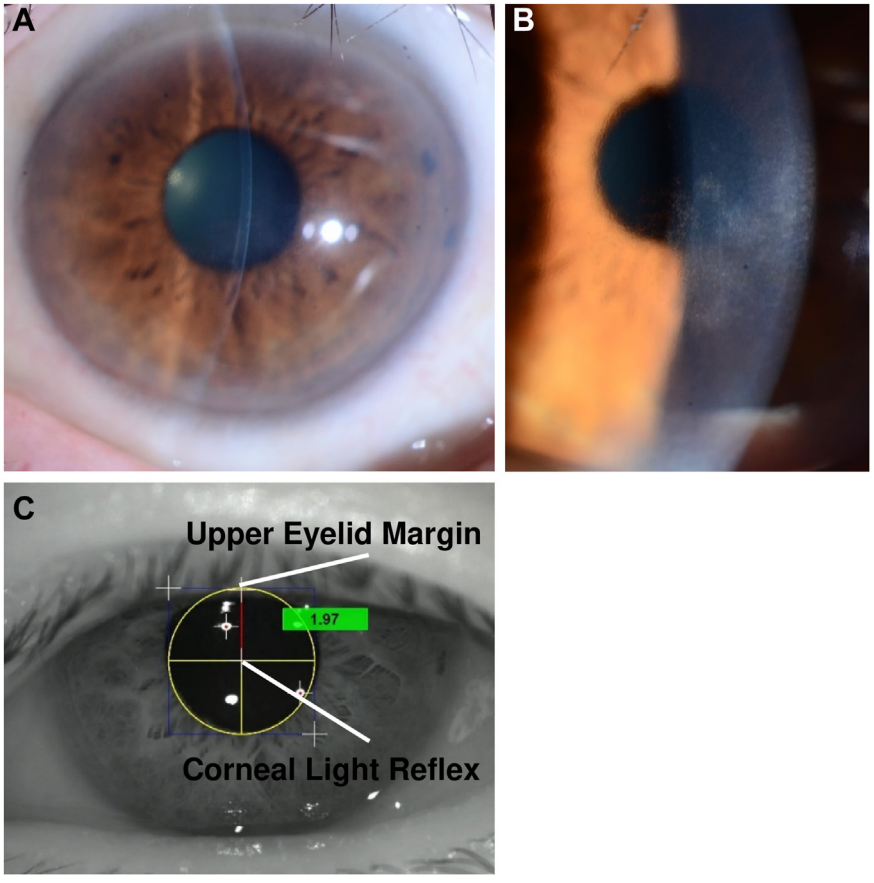


Figure 1. Representative image of a 66-year-old man with FECD. **A,** Slit-lamp microscopy image. **B,** Confluent guttae classified as grade 5 according to the modified Krachmer grading scale. **C,** Measurement of MRD using Idra. FECD = Fuchs endothelial corneal dystrophy; MRD = margin reflex distance.

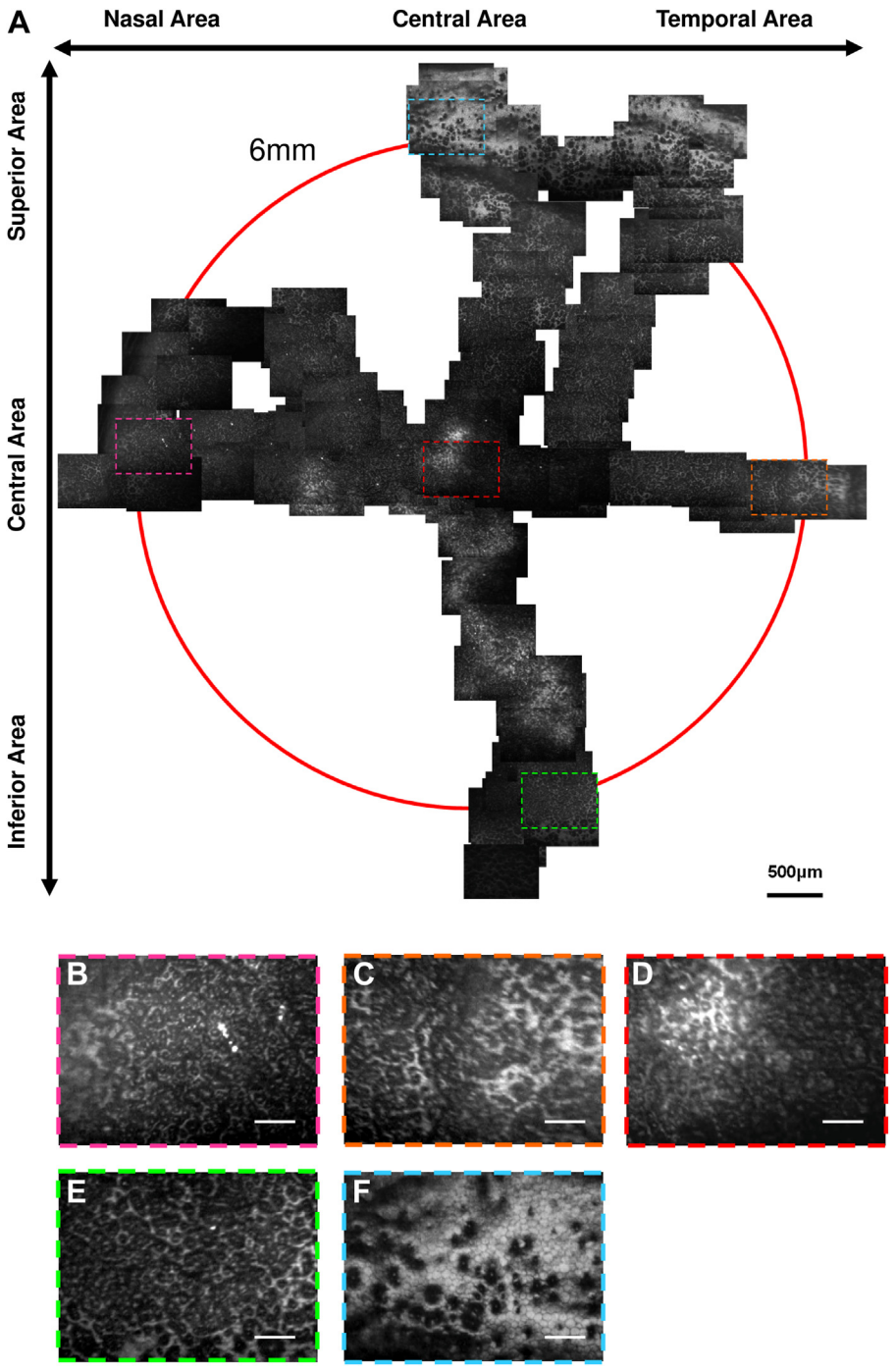


Figure 2. Analysis of CEC distribution using slit-scanning wide-field contact specular microscopy in a 66-year-old man with FECD. **A,** Comprehensive panoramic imaging with extraction points from 5 distinct regions: the central area and 4 peripheral locations at 6 mm diameter intervals from the center. Scale bar = 500 μm. **B–F,** Representative panoramic images from: **(B)** nasal area; **(C)** temporal area; **(D)** central area; **(E)** inferior area; **(F)** superior area. Scale bar = 100 μm. CEC = corneal endothelial cell; FECD = Fuchs endothelial corneal dystrophy.

and identification of guttae and dark band regions, which indicated extensive extracellular matrix presence. The identification of dark regions was accomplished by iteratively applying the minimum extended minima operator within the MorphoLibJ library.[19] After automated detection, an ophthalmologist conducted a manual review to exclude artifacts, such as extensive shadowing caused by corneal endothelial tissue folds, and to isolate regions of suspected abnormality. All processed images were subsequently verified by a senior corneal specialist in a masked fashion. The ratio between guttae areas and total CEC regions was calculated by comparing

the original images with those extracted through mathematical morphological processing (Fig 3A, B).

Analysis of Guttae Distribution in Relation to Eyelid Position

Superior images were extracted at 3 mm from the central region. To investigate the relationship between eyelid position and superior guttae distribution, patients were divided into 2 groups based on MRD: group A with MRD ≤3 mm and group B with MRD >3 mm. The percentage of guttae of the superior area was analyzed for each group.

Statistical Analysis

Statistical analyses were performed using Prism 10 software with 10.0.2 (GraphPad Software) statistical analysis software. The analysis of guttae area was performed using 1-way analysis of variance followed by Tukey multiple comparison test. Comparisons between MRD groups were performed using the Mann–Whitney *U* test.

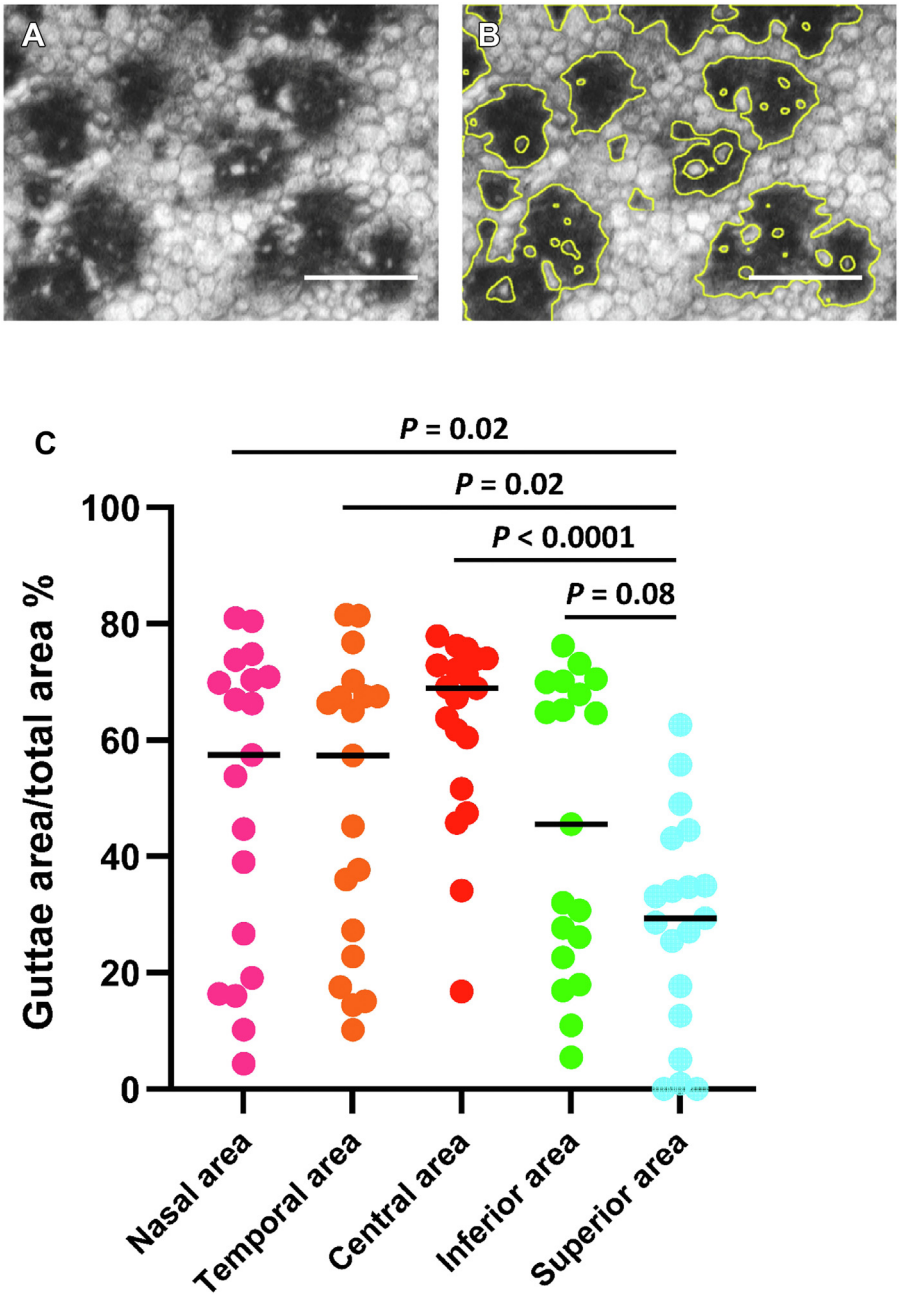


Figure 3. Quantitative analysis of guttae distribution using custom image analysis. **A** and **B,** Representative images showing automated guttae boundary detection (yellow lines indicate boundaries of extracted abnormal regions). **C,** Regional comparison of proportion of guttae, calculated as the percentage of total image area occupied by guttae in each region. Scale bar = 100 μm.

Table 1. Patient Characteristics and Regional Guttae Distribution Analysis

Age and Sex	Eye	Margin Reflex Distance (mm)	Modified Krachmer Grade	Percentages of Guttae				
				Nasal Area (%)	*Temporal Area (%)*	*Central Area (%)*	*Inferior Area (%)*	*Superior Area (%)*
75, F	L	2.21	4	16.34	10.22	51.62	10.95	0.94
67, F	R	3.47	5	44.71	22.83	34.13	27.72	33.12
68, F	R	2.43	5	57.48	57.36	61.71	65.21	62.61
72, M	L	2.48	5	70.88	36.08	60.47	22.61	34.93
82, F	R	2.25	5	69.91	76.81	68.98	69.93	49.00
47, F	R	2.96	5	10.22	15.15	76.17	5.48	0
85, M	R	3.15	5	26.72	65.00	71.19	45.57	27.03
74, F	L	2.82	5	16.01	37.74	69.12	32.01	0
81, M	L	3.69	5	73.81	81.41	72.86	64.63	44.55
81, F	R	2.59	5	4.34	45.19	73.94	30.67	25.46
80, F	R	3.01	5	74.76	67.53	75.72	64.76	34.72
74, M	R	2.65	5	70.38	67.40	77.90	70.15	43.16
53, F	R	3.7	5	66.31	81.55	72.20	76.19	55.76
64, F	L	3.69	4	39.01	66.40	45.75	73.10	34.08
65, F	R	2.4	4	19.15	27.27	16.75	16.95	5.10
72, M	L	2.69	3	80.93	17.57	47.42	17.99	29.39
56, F	R	3.99	5	53.83	14.49	67.28	26.14	12.59
55, F	R	2.38	5	66.95	70.20	63.77	70.58	17.73
66, M	L	1.97	5	80.44	67.57	74.00	67.85	28.60

F = female; L = left; M = male; R = right.

Results

Patients

The mean age of the patients at the time of analysis was 72 years old, with ages ranging from 47 to 85 years old; the group included 6 men (31.6%) and 13 women (68.4%), with 12 right eyes (63.2%) and 7 left eyes (36.8%). The grade of corneal guttae was classified based on modified Krachmer,[20] which revealed 1 eye in grade 3, 3 eyes in grade 4, and 15 eyes in grade 5. All subjects were Japanese, and the study did not include patients from different ethnic backgrounds. Patient characteristics and relevant clinical data, including MRD, are summarized in Table 1.

Percentage of Guttae in the Different Areas

The median percentages of guttae in the central, superior, temporal, nasal, and inferior areas were 69.0 (95% confidence interval [CI], 60.4%−77.5%), 29.4 (95% CI, 19.6%−39.2%), 57.4 (95% CI, 44.2%−70.5%), 57.5 (95% CI, 44.6%−71.3%), and 45.6 (95% CI, 32.5%−58.6%), respectively. There were significant differences between the superior area and central area, superior area and temporal area, and superior area and nasal area, but there were no significant differences between the superior area and inferior area. This reveals that the superior area had significantly fewer guttae compared with other corneal areas (Fig 3C).

Comparison of Superior Guttae Distribution between 2 Groups

The study population was divided into 2 groups: 13 eyes in group A with MRD ≤3 mm and 6 eyes in group B with MRD >3 mm. The median percentage of superior guttae was 28.6 (95% CI, 15.7%−41.5%) in group A and 33.6 (95% CI, 18.9%−48.3%) in group B, and there were no significant differences between the groups ($P = 0.40$) (Fig 4).

Discussion

In this study, we assessed the distribution of guttae in 5 distinct corneal regions using slit-scanning wide-field

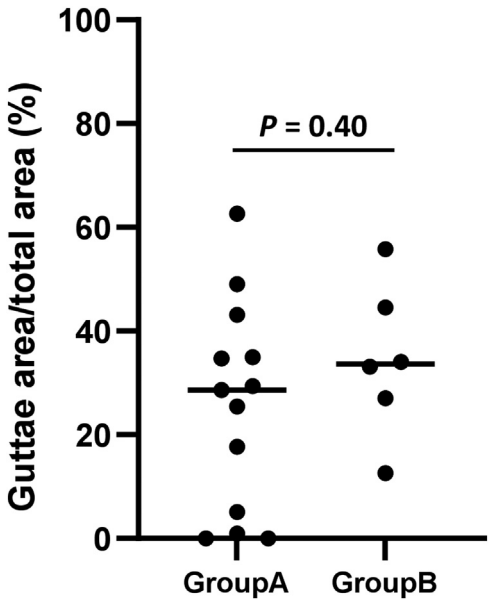


Figure 4. The study population was divided into 2 groups: group A with MRD ≤3 mm and group B with MRD >3 mm. The percentage of guttae of the superior area was analyzed for each group. MRD = margin reflex distance.

contact specular microscope images of patients with FECD. Our analysis revealed that the superior corneal region demonstrated a significantly lower proportion of guttae compared with other regions, particularly the central area. This finding suggests a potential protective effect, although the correlation between eyelid position and guttae formation was not statistically significant when analyzed through MRD measurements.

While the traditional understanding suggests a central predominance of guttae with peripheral sparing in FECD,[1] our quantitative analysis revealed a more nuanced distribution pattern. The highest density of guttae was observed in the central region, with substantial involvement of the temporal and nasal regions, whereas the superior region showed significantly lower density. Moreover, the inferior cornea, which has some protection by the lower eyelid, showed the trend toward a lower proportion of guttae compared with the central, temporal, and nasal regions, leading to a nonsignificant difference between inferior and superior portions. This distribution pattern notably corresponds to the anatomical configuration of the human eye, specifically the horizontal elliptical shape of the palpebral fissure. Recent studies have demonstrated that oxidative stress induced by UV-A irradiation plays a crucial role in the pathogenesis of FECD.[9,21] Similarly, the biomechanical properties of the cornea cause a reduction in UV light transmission in the peripheral cornea compared with the center, leading to the most prominent radiation damage in the central CECs.[22] The increased presence of guttae along the horizontal meridian—areas consistently exposed due to minimal eyelid coverage—suggests a potential relationship between corneal exposure and guttae formation. Additionally, we observed minimal differences between the temporal and nasal regions. Although Coroneo et al has demonstrated that peripheral light focusing concentrates light nasally,[23,24] recent research suggests the contribution of peripheral light focusing to direct UV exposure is limited.[25] Furthermore, Coroneo et al's peripheral light focusing effect in pterygium primarily applies to UV-B radiation, whereas most UV radiation that penetrates to the corneal endothelium is UV-A, and the difference in UV-A exposure between temporal and nasal regions at the innermost layer of the cornea remains unknown.[22] These factors may explain why no significant difference was observed in guttae distribution between the temporal and nasal regions.

These findings raise important clinical implications, particularly regarding preventive strategies. The significantly lower proportion of guttae in the superior cornea suggests that UV protection might be beneficial in early FECD. In our analysis, superior corneal images were extracted at 3 mm from the center, and patients were stratified into 2 groups based on an MRD threshold of 3 mm. However, no significant differences were observed between these groups. This lack of significant correlation may be attributed to several factors. First, as FECD progresses, guttae tend to become confluent throughout the cornea, potentially masking regional differences between the superior and central areas. In our study, the majority of patients (15 of 19 eyes) were classified as grade 5 according to the modified Krachmer classification, representing advanced disease. This predominance of severe cases might explain why we did not observe significant differences between MRD groups, as extensive guttae formation in advanced disease could mask subtle regional variations that might be more apparent in earlier stages. Additionally, the complex pathogenesis of FECD, involving multiple genetic and environmental factors, may contribute to this variability in guttae distribution patterns.[1]

Our findings both support and extend previous observations that reported regional variations in guttae distribution using noncontact specular microscopy. However, our use of slit-scanning wide-field contact specular microscopy provided higher-resolution images and a more detailed analysis of guttae distribution, particularly in cases with confluent guttae where noncontact methods have limitations. This methodological advancement allows for more precise quantification of regional differences and may better inform our understanding of disease progression patterns.

Several limitations of our study should be noted. First, our analysis was limited by the relatively small number of eyes enrolled. The observed difference between superior and inferior corneal regions did not reach statistical significance ($P = 0.08$), which may be attributed to the small cohort size (only 19 eyes from 19 patients). This limitation potentially affected our ability to detect subtle differences between these regions. Second, we did not assess the genetic profile of the patients, particularly regarding known FECD-associated mutations, which could potentially influence the regional distribution patterns of guttae. Third, although we examined 5 principal corneal regions (central, superior, temporal, nasal, and inferior), our analysis did not include all 9 corneal zones, thus potentially missing subtle variations in the overall corneal distribution pattern of guttae. Finally, we were only able to evaluate up to 6 mm from the center, as the peripheral areas could not be properly contacted with the cone lens of the contact specular microscope, leaving the far peripheral guttae distribution unevaluated.

In conclusion, our study demonstrated distinct regional variations in guttae distribution in patients with FECD, with lower density in the superior corneal region compared with other areas, with statistically significant differences found for nasal, temporal, and central comparisons ($P < 0.05$) but not for the inferior area ($P = 0.08$). While the correlation between eyelid position and guttae formation was not statistically significant, the observed distribution pattern, particularly along the horizontal meridian, suggests a potential relationship between environmental exposure and guttae formation. Despite the limitations of our study, including sample size and genetic profiling, our observations provide valuable insights that may inform future preventive strategies, such as UV protection, for patients with FECD. Further large-scale studies incorporating genetic analysis and more comprehensive corneal mapping are warranted to fully elucidate the complex interplay between anatomical, genetic, and environmental factors in FECD progression.

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HUMAN SUBJECTS: Human subjects were included in this study. This prospective observational study was approved by the Ethics Committee of Kyoto Prefectural University of Medicine (KPUM), Kyoto, Japan, and was conducted in accordance with the principles of the Declaration of Helsinki. Patients were enrolled after obtaining informed consent.

No animal subjects were used in this study.

Author Contributions:

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Data collection: Tomioka, Yamamoto, Kitazawa, Fukuoka

Analysis and interpretation: Kinoshita, Tomioka, Yamamoto, Tanaka, Sotozono, Jurkunas

Obtained funding: N/A

Overall responsibility: Kinoshita, Tomioka, Ueno, Yamamoto, Sotozono, Jurkunas

Abbreviations and Acronyms:
CEC = corneal endothelial cell; **CI** = confidence interval; **FECD** = Fuchs endothelial corneal dystrophy; **MRD** = marginal reflex distance; **UV** = ultraviolet.

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